Bot Hunter Analysis Report

1. Executive Summary & Problem Statement

Bot Hunter analyses fictitious ad-click traffic to identify events that look more like automated bot activity than legitimate user behaviour. The business challenge is a familiar one for advertising platforms: invalid clicks can inflate campaign metrics, distort billing, reduce trust in performance reporting, and consume operational time that should be spent on genuine customer outcomes.

This run read `data/bot-hunter-dataset.tsv` and analysed 149,239 click events. Each row contains an event identifier, timestamp, region, browser, operating system, and click URL. The URL query string carries behaviour signals such as clicked domain (`d`), search query (`q`), time to click (`ttc`), and country-like context (`ct`).

Bot Hunter selected 3,731 events, or 2.50% of the batch, as likely bot traffic for review. This is a meaningful operational finding: even a small invalid-click rate can affect revenue assurance, customer confidence, campaign optimisation, and the credibility of downstream reporting.

This report should be read as evidence for review, not as proof of fraud for every individual event. The dataset does not include ground-truth labels, so the system cannot claim measured precision, measured recall, or a calibrated fraud probability. The submitted output is `submission.tsv`. It keeps the required binary `is_bot` decision and adds `operational_tier` so the same prediction can be used carefully in business workflows.

2. Methodology & Rationale

Bot Hunter uses two classifiers because the risk has two different shapes. A rules classifier catches patterns that are easy to explain and audit, such as repeated query/domain pairs, mechanical timing, and dense bursts. The Extended Isolation Forest model catches unusual combinations across engineered features that fixed rules may miss.

This combined design was chosen over a simpler single-rule approach because a single rule is too brittle for adversarial traffic. For example, a bot can avoid one obvious threshold while still leaving a broader statistical footprint across timing, repeated terms, clicked domains, and device-like context. The rules layer gives leadership an auditable explanation; the anomaly model adds coverage where fixed rules would otherwise be narrow.

Component	Current choice	Why this choice was made
Rules classifier	Weighted heuristic rules	Transparent, auditable, and suitable for explaining repeated or mechanical click patterns.
ML classifier	Extended Isolation Forest model	Works without labels and can detect unusual combinations across engineered features.
Score blend	`0.58 * heuristic_score + 0.42 * ml_score`	Keeps explainable rule evidence dominant while still using the anomaly model.
Main cutoff	97.5th percentile combined-score threshold	Keeps the selected population stable for an unlabelled batch.
Override	`heuristic_score >= 0.62`	Protects high-confidence rule evidence from being missed by model ranking shifts.

The final score is a weighted blend:

```
combined_score = (0.58 * heuristic_score) + (0.42 * ml_score)
```

The rules layer is weighted slightly higher because it is easier to explain and audit. The anomaly model still matters because it can identify unusual combinations that a small rule set may not capture.

Raw URLs are not model-ready. Bot Hunter parses each query string and converts it into behavioural features. Count features are log-transformed because click-log counts are heavy-tailed: a few domains or queries may appear many times while most appear once. `kp` and `sld` are treated as categorical identifiers, not ordered numeric measurements. Their raw values remain in `artifacts/features.tsv` for audit, but the model uses `log_kp_count` and `log_sld_count` to learn concentration without inventing false numeric distance between categories.

The rules layer currently scores:

- repeated query/domain pairs
- repeated queries
- confirmed query repetition
- high-volume clicked domains
- dense region/browser/OS clusters
- exact time-to-click reuse
- same-second bursts
- dense burst repetition clusters
- concentrated `ct` context paired with repetition and clustering
- implausibly fast clicks
- moderately long or extreme time-to-click values
- regular pseudo-session inter-arrival timing

Rule contributions are separated into strong and supporting evidence. Strong rules represent direct mechanical or replay-like patterns, such as impossible timing or repeated query/domain behaviour. Supporting rules provide context, such as volume, concentration, or broad clustering. Supporting contributions are capped so several weak signals cannot add up to the same meaning as a strong bot signal on their own.

Rule strength	Meaning	Score treatment
Strong	Direct mechanical or replay evidence	Contributes its full rule weight
Supporting	Contextual or weaker evidence	Capped together at 0.24

The exact time-to-click reuse rule uses a 99th-percentile reuse-count threshold with an absolute floor. The main repetition and concentration rules also use 99th-percentile batch thresholds with the previous fixed count and total-rate cutoffs kept as guardrails. This lets the rules adapt to the observed population while avoiding weak duplicate counts in small runs.

The `ct` rule, confirmed query repetition, and dense burst repetition rules are conjunctive. In practical terms, they require multiple signals to be present before adding score. This is safer than treating broad context, such as country-like concentration, as a standalone bot indicator.

The anomaly classifier uses the engineered feature matrix. The main feature families are:

- region/browser/OS frequency
- global `ct` country frequency
- sub-200 ms click flags
- local burst density
- query entropy
- query, domain, and query/domain repetition counts
- same-second and exact time-to-click reuse counts
- timing magnitude after log transformation
- low-cardinality `kp` and `sld` value frequencies

High-volume domain frequency and global country frequency are down-weighted to 0.50 and 0.50, respectively. The low-cardinality `kp` and `sld` frequency features are down-weighted to 0.50 and 0.25. Events isolated quickly by random hyperplane splits receive higher anomaly scores. EIF is the only production anomaly model; alternate ML backends and supervised pilots have been removed.

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3. Core Statistical Findings

The current run shows a concentrated anomaly population rather than a broad quality problem across the whole dataset. Bot Hunter selected 3,731 of 149,239 events as likely bot traffic. That is 2.50% of the run, leaving the large majority of traffic in the monitor population.

Metric	Current value	Plain-English meaning
Total events analysed	149,239	Number of click events in the batch.
Selected bot events	3,731	Events marked `is_bot = 1`.
Selected bot rate	2.50%	Share of the batch selected as likely bot traffic.
Suppress tier	1,863	Strongest operational candidates, subject to policy approval.
Quarantine tier	1,868	Suspicious events that should be reviewed or sampled.
Monitor tier	145,508	Events not selected for bot action.
Combined-score cutoff	0.568053	Run-specific anomaly threshold.
Heuristic-only flag rate	1.36%	Share flagged by rules before model blending.
ML agreement-tail rate	2.50%	Reference tail used to compare ML agreement.
Operational confidence estimate	74.49%	Signal-based estimate, not measured precision.

Top explainable rule patterns from the current run:

Pattern	Events	How to explain it
repeated query	3,403	The same search term appears unusually often.
repeated query/domain pair	2,086	The same search term repeatedly clicks the same domain.
confirmed query repetition	2,023	Query repetition is backed by additional context.
very short query	1,798	Short terms can be legitimate, but are useful supporting evidence.
high-volume clicked domain	1,732	A domain receives unusually concentrated clicks.
heavy region/browser/OS cluster	1,168	Traffic is concentrated in a narrow device/server footprint.
concentrated ct context	904	Country-like context is concentrated with repetition and clustering.
same-second click burst	677	Multiple clicks happen in the same second.
moderately long time-to-click	431	Timing is unusual enough to support other evidence.
dense burst repetition cluster	236	Burstiness and repeated behaviour appear together.

Method agreement from the current run:

Bucket	Events	Interpretation
Heuristic + ML	1,726	Strongest review candidates because both methods agree.
Heuristic only	311	Explainable rule hits that are not extreme in the ML feature space.
ML only	2,005	Multivariate anomalies that need careful sampling or review.
Neither strong	145,197	Traffic not strongly indicated by either method.

`Combined tail` is the borderline evidence bucket. It means the event was selected because the blended `combined_score` crossed the run threshold, while neither strong rules nor the strongest ML-only evidence was enough on its own. Treat these events as combined evidence for review or quarantine, not as automatic suppression candidates.

Operational anomaly classes from the current run:

Class	Selected events	Recommended handling
Repetition with supporting context	1,877	Review as replay-like traffic. Suppress only when the event is already in the suppress tier; otherwise quarantine or sample.
Compound burst/replay	562	Treat suppress-tier events as strong operational candidates; quarantine the rest for timing-pattern review.
ML-tail multivariate anomaly	509	Quarantine or sample. Do not suppress automatically without feature-deviation review or labels.
Repetition with timing anomaly	355	Prioritise when suppress-tier or when timing evidence is strong; otherwise quarantine for sampling.
Repetition dominated	350	Use as an explainable replay candidate. Quarantine lower-score cases when ML agreement is absent.
Supporting context plus combined tail	77	Monitor or quarantine. These are useful for trend review, not standalone suppression.
Other combined-tail anomaly	1	Sample manually before taking action.

The largest pattern is repetition with supporting context. In practical terms, that means the same queries and clicked domains appear far more often than expected and are reinforced by other signals such as device-like clustering, country-like context, or timing behaviour. The ML-only population is also material, but it carries a different business meaning: those events are unusual in the feature space and should be sampled or quarantined rather than treated as automatically fraudulent.

4. Anomaly Explanations & Practical Guidance

Repetition is suspicious when the same query, clicked domain, and device-like context appear together. Timing matters because some click patterns are difficult for humans to produce consistently. Country-like `ct` concentration is supporting evidence only; it should not be treated as a country-blocking rule. ML-only events are unusual, but unusual does not automatically mean fraudulent.

Operational anomaly classes derived from the current unlabelled run; these are not proven fraud labels.

The classes below group the 3,731 selected events from this run. The grouping is backed by full-run rule contributions, rule strength and family fields, heuristic and ML scores, method-agreement buckets, operational tiers, and the run-specific thresholds described later in this report.

<table>
<tr><th>Class</th><th>Selected events</th><th>Data backing</th></tr>
<tr><td>Repetition with supporting context</td><td>1,877</td><td>tiers
suppress 1,327
quarantine 550
methods
Heuristic + ML 1,198
Heuristic only 186
Combined tail 493
top rules
repeat_query (1,877)
repeat_query_domain (1,384)
confirmed_query_repetition (1,384)</td></tr>
<tr><td>Compound burst/replay</td><td>562</td><td>tiers
suppress 154
quarantine 408
methods
Heuristic + ML 150
Heuristic only 11
Combined tail 401
top rules
repeat_query (558)
same_second_burst (441)
short_query (285)</td></tr>
<tr><td>ML-tail multivariate anomaly</td><td>509</td><td>tiers
quarantine 509
methods
ML only 509
top rules
This class is grouped by anomaly-model agreement, not by rule explanations. Low-weight rules may be present, but they do not cross the heuristic override threshold.</td></tr>
<tr><td>Repetition with timing anomaly</td><td>355</td><td>tiers
suppress 199
quarantine 156
methods
Heuristic + ML 195
Heuristic only 4
Combined tail 156
top rules
repeat_query (355)
moderate_long_ttc (283)
repeat_query_domain (199)</td></tr>
</table>

<table>
<tr><th>Class</th><th>Selected events</th><th>Data backing</th></tr>
<tr><td>Repetition dominated</td><td>350</td><td>tiers
suppress 183
quarantine 167
methods
Heuristic + ML 183
Heuristic only 110
Combined tail 57
top rules
repeat_query_domain (350)
repeat_query (293)
confirmed_query_repetition (293)</td></tr>
<tr><td>Supporting context plus combined tail</td><td>77</td><td>tiers
quarantine 77
methods
Combined tail 77
top rules
same_second_burst (74)
high_volume_domain (62)
heavy_device_cluster (62)</td></tr>
<tr><td>Other combined-tail anomaly</td><td>1</td><td>tiers
quarantine 1
methods
Combined tail 1
top rules
same_second_burst (1)
fast_click (1)</td></tr>
</table>

The ML-tail population contains 2,005 events with high anomaly scores and heuristic scores below the rule override threshold. Only the subset that also crossed the combined-score cutoff is counted as selected traffic in the class table. This keeps ML-only anomalies visible without describing them as rule-derived replay evidence.

Concrete examples from the current run are summarised below. The underlying event-level fields, scores, tiers, and rule evidence can be visualised in the Traffic Explorer in the dashboard.

- Repetition with supporting context: This is a replay-like example rather than a single odd click. The same query/domain pattern around `nomnem` and `www.amazon.de` is reinforced by broader context such as domain volume, device-like clustering, and country-like concentration. That combination is stronger than repetition alone because several independent fields point in the same direction. The example event is `evt_147679`, which clicked `www.amazon.de` for query `nomnem`.
- Compound burst/replay: This example shows repetition happening inside a burst. The key concern is not only that the query and domain repeat, but that the click pattern also appears in same-second or dense burst evidence. That is more consistent with automated replay than with ordinary human browsing. The example event is `evt_097085`, which clicked `duckduckgo.com` for query `vielle motoneuron`.
- ML-tail multivariate anomaly (2,005 events in the wider population): This is an unusual multivariate event rather than a clean rule hit. The rules did not provide enough strong evidence for automatic suppression, but the event sits far enough into the anomaly-model tail to justify quarantine or sampling. In business terms, this is a review candidate, not proof of fraud. The example event is `evt_082067`, which clicked `www.firefold.com` for query `splenocyte`.
- Repetition with timing anomaly: This example combines repeated query/domain behaviour with unusual timing. Repetition explains why the event resembles replay; the timing evidence makes the case stronger because the click cadence does not look like normal human variation. The example event is `evt_004943`, which clicked `www.amazon.ca` for query `nomnem`.
- Repetition dominated: This is the simplest explainable replay pattern. The main evidence is repeated query/domain behaviour with confirmed query repetition, without needing burst or broader context to carry the decision. The example event is `evt_009777`, which clicked `www.overstock.com` for query `fuzz sulphid`.
- Supporting context plus combined tail: This example is deliberately treated as borderline. It has supporting context and a high enough blended score to cross the run threshold, but neither the rules nor the ML evidence is strong enough on its own. That is why the right action is review or quarantine rather than automatic suppression. The example event is `evt_089535`, which clicked `www.amazon.co.uk` for query `aw`.
- Other combined-tail anomaly: This is a residual combined-tail example. It crosses the blended threshold, but it does not match the main operational classes. The fast-click and burst evidence make it worth sampling, while the borderline score argues against automatic suppression. The example event is `evt_032590`, which clicked `find.gmx.com` for query `balm thyrohyal youre eight`.

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5. Recommended Business Actions

Separate prediction from action. `is_bot` is the compatibility prediction; `operational_tier` is the safer business-control layer.

Tier	Recommended action	Business assumption
`suppress`	Exclude from billing or downstream metrics after policy approval.	The organisation accepts limited review risk for high-confidence traffic.
`quarantine`	Hold, delay, sample, or manually review before suppression.	Ambiguous traffic should not be automatically removed.
`monitor`	Keep for trend monitoring and future labels.	Non-selected traffic can still help detect drift later.

Operational split in this run:

- suppress: 1,863 events
- quarantine: 1,868 events
- monitor: 145,508 events

Use `suppress` for the strongest operational candidates after policy approval. Use `quarantine` as the default action for ambiguous or ML-only traffic. Do not automatically block traffic solely because it is in the ML tail.

These tiers are the recommended operating model because they keep action proportionate to the strength of evidence. Teams that need to go beyond the three-tier view can also use the more targeted filters below. Treat them as review controls for similar unlabelled datasets, not as ground-truth fraud rules.

Practical filtering options for similar unlabelled datasets:

Filter	Use	Caveat
Conservative suppression review: `operational_tier == 'suppress'`	Start here for the strongest operational candidates. Still requires policy approval because labels are unavailable.	Check policy, billing, and customer-impact rules before action.
Quarantine for manual review: `operational_tier == 'quarantine'`	Hold, sample, or delay action on suspicious traffic that is not strong enough for direct suppression.	Use sampling to estimate likely false positives before suppression.
Explainable replay review: `anomaly_class in repetition_with_supporting_context, compound_burst_replay, repetition_with_timing, repetition_dominated`	Focus reviewer time on repeated query/domain behaviour with clear rule evidence.	Validate repeated-pattern assumptions against campaign context.
ML-tail sampling: `ml_score >= 0.975 and heuristic_score < 0.62`	Sample for future feature-deviation work; do not treat as proven fraud without labels.	Needs feature-deviation review because rule evidence is below override.

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6. Probability Perspective & Risk Assessment

The current operational confidence estimate is 74.49%. This is not measured precision because the dataset has no ground-truth labels.

Confidence is higher when independent signals agree: for example, when a strong repeated query/domain rule hit also lands in the upper anomaly-model tail. Confidence is lower for ML-only events because anomaly detection can also surface legitimate edge cases such as unusual campaigns, regional spikes, testing traffic, or rare but valid user behaviour.

The estimate is based on rule/model agreement, strength of rule evidence, operational tier, selected anomaly class, and the known absence of labels. It should guide review priority, not replace labelled validation or policy approval.

Risk area	Likelihood	Impact	Business interpretation
Suppress-tier bot traffic remains untreated	High	Medium to High	Strong signals would continue to distort billing, reporting, and optimisation unless policy-approved action is taken.
Quarantine-tier traffic is suppressed too aggressively	Medium	Medium	Ambiguous or ML-only events may include legitimate edge cases, so sampling and review should precede irreversible action.
Model drift in future batches	Medium	Medium	Traffic mix, campaigns, browsers, and geographies can change; thresholds should be monitored rather than assumed permanent.
Treating anomaly scores as proof of fraud	Low if governed	High	The current scores are evidence for review, not legal or contractual proof. Misuse would create customer and reputational risk.

If no action is taken, the likely exposure is continued metric inflation and avoidable investigation effort. The current evidence does not quantify direct financial loss, but it does show a reviewable population large enough to affect performance reporting if left unmanaged.

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7. Generalisation, Trade-Offs, And Limitations

The approach should generalise to bot traffic that is repetitive, mechanically timed, or concentrated in narrow device and query patterns. It may miss bots that deliberately randomise timing, distribute activity across many query/domain pairs, or closely imitate human behaviour.

The main trade-off is explainability versus coverage. Rules are easier to defend but can miss novel patterns. The anomaly model broadens coverage but needs careful interpretation because it finds unusual events, not necessarily fraudulent events.

Known limitations:

- There are no ground-truth labels, so measured precision, recall, and calibration cannot be reported.
- Legitimate campaigns can create high repetition, dense clusters, or regional concentration.
- The `ct` rule is supporting evidence only; it should not be used as a country-level blocking rule.
- The regular inter-arrival rule uses pseudo-session groups because there is no explicit user or session identifier.
- The anomaly score is relative to the current batch and should be monitored for drift across future runs.

Material changes in geography, campaign volume, inventory, browser mix, or feature extraction should trigger fresh threshold review.

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8. Future Work

The roadmap should improve the system in three directions: make the current signals easier to explain, make the thresholds more stable across batches, and replace estimated confidence with measured performance once trusted labels exist.

Near-term work should focus on improvements that do not require external credentials or labelled data:

1. Add feature-deviation explanations for ML-tail events. A reviewer should see why an ML-only event was unusual, for example that its query/domain repetition or same-second density sits in the top 1% of the batch.
2. Add optional local domain reputation signals. A versioned local reputation file can add useful context without making live network calls or turning a broad reputation match into an automatic bot decision.
3. Compare robust scaling or quantile transforms for heavy-tailed features. This would test whether the model is too sensitive to very large domain, query, or device-cluster counts.

Medium-term work should make the approach safer across changing traffic mixes:

1. Normalise high-volume signals by available context such as region, browser, operating system, country-like `ct`, hour, and future campaign or inventory metadata. High volume is more suspicious when it is concentrated in a narrow footprint than when it is broad and expected.
2. Add rolling burst features over 1-second, 10-second, and 60-second windows. This would catch automation that avoids exact same-second bursts by spacing clicks just far enough apart.
3. Store compact run history for drift monitoring, including flagged rate, score quantiles, tier counts, top reasons, and top domains. A sudden move from a 2.5% selected rate to a much higher rate should be visible before the output is treated as normal.

Longer-term work needs more operational context:

1. Add cached live reputation providers only when credentials, usage terms, and data-handling requirements are clear. Live lookups should be optional, cached by unique domain, and disabled by default.
2. Collect labelled validation from manual review, invalid-traffic feedback, chargebacks, confirmed abuse reports, or trusted campaign investigations. Labels are required before reporting measured precision, recall, calibration, or optimised decision thresholds.
3. Build a feedback loop from reviewed `suppress` and `quarantine` decisions. Once label quality is good enough, the team can evaluate supervised models while keeping the current rules plus Extended Isolation Forest path as the production baseline.

Appendix A: Metric Definitions

Metric	Definition
`is_bot`	Required binary prediction in `submission.tsv`; `1` means selected as likely bot traffic.
`heuristic_score`	Score from transparent rules such as repetition, timing, and burst checks.
`ml_score`	Anomaly score from the Extended Isolation Forest feature model.
`combined_score`	Weighted blend of rule and ML evidence.
Combined-score cutoff	Run-specific threshold used to select the top anomaly population.
Heuristic override	Rule-score threshold that selects strong explainable rule hits.
Operational tier	Business action layer: `suppress`, `quarantine`, or `monitor`.
Operational confidence estimate	Signal-based estimate from method agreement; not measured precision.
Method bucket	Agreement category showing whether rules, ML, both, or neither were strong.
`Combined tail`	Borderline method bucket selected by the blended `combined_score`; neither strong rules nor strongest ML-only evidence was enough alone, so use it for review or quarantine rather than automatic suppression.
Anomaly class	Human-readable grouping of selected events by dominant evidence pattern.

Appendix B: Feature Definitions

Feature	Meaning
`log_domain_count`	Log-scaled frequency of the clicked domain.
`log_query_count`	Log-scaled frequency of the search query.
`log_query_domain_count`	Log-scaled frequency of the query/domain pair.
`log_device_count`	Log-scaled frequency of the region/browser/OS cluster.
`log_country_count`	Log-scaled frequency of `ct`, the country-like URL parameter.
`log_same_second_count`	Log-scaled number of clicks sharing the same second.
`log_ttc_count`	Log-scaled reuse count for exact time-to-click values.
`log_kp_count`	Log-scaled frequency of the `kp` URL parameter value.
`log_sld_count`	Log-scaled frequency of the `sld` URL parameter value.
`kp`	Raw `kp` URL parameter retained for audit; excluded from ML features.
`sld`	Raw `sld` URL parameter retained for audit; excluded from ML features.
`hour`	Event hour extracted from the timestamp.
`log_ttc_seconds`	Log-scaled time-to-click duration.
`is_sub_200ms_click`	Flag for click timing below a plausible human reaction threshold.
`log_pseudo_session_10s_click_count`	Log-scaled burst density within pseudo-session-style groups.
`query_entropy`	Text randomness measure for the search query.

Appendix C: Model Definition

Extended Isolation Forest is an unsupervised anomaly-detection model. It does not learn from known bot labels. Instead, it asks which events are easier to isolate from the rest of the batch using random feature-space splits.

In plain English, events that look very different from the main population tend to be isolated more quickly and receive higher anomaly scores. This is useful for an unlabelled dataset, but it also means the model detects unusual behaviour rather than confirmed fraud. The model output is strongest when it agrees with transparent rule evidence and weakest when it stands alone without feature-deviation review.

Methods evaluated but not selected:

Method	Why it was not preferred
K-means clustering	K-means is useful when compact, roughly spherical clusters are expected. Bot-click traffic is sparse, skewed, and heavy-tailed, so forcing every event into a cluster makes the output harder to explain and less reliable for rare anomalies.
Standard Isolation Forest	Standard Isolation Forest is a strong baseline, but its axis-aligned splits can be less expressive when anomalous behaviour is a combination of several weak signals. Extended Isolation Forest uses random hyperplane splits, which better matches the multivariate footprint of repeated, bursty, and concentrated traffic.
DBSCAN	DBSCAN can find dense groups and outliers, but it is sensitive to distance scaling and neighbourhood parameters. With mixed count, timing, entropy, and categorical-frequency features, it was less stable as a default production choice for repeatable local batch runs.

Extended Isolation Forest was preferred because it preserves the main advantage of unsupervised Isolation Forest-style detection while giving a more flexible view of multivariate anomalies. It also fits the operating constraints of this project: no labels, a local batch pipeline, explainable feature engineering, and a need to combine model scores with transparent rules.

Appendix D: Thresholds And Decision Logic

An event is selected as a bot when either condition is true:

```
combined_score > 0.568053
or heuristic_score >= 0.62
```

The combined-score threshold is the run-specific 97.5th-percentile cutoff. It keeps the selected volume stable for an unlabelled dataset while letting the ML model influence borderline cases.

The heuristic override protects high-confidence, explainable rule hits. For example, a strongly repeated query/domain pattern with mechanical timing should not be missed only because the anomaly ranking moved after a feature change.

Adaptive heuristic thresholds used in this run:

Rule	Computed threshold	Basis
Concentrated ct context (`concentrated_ct_context`)	29013	99th-percentile threshold, floor 1000, rate guardrail 14923
Heavy region/browser/OS cluster (`heavy_device_cluster`)	43674	99th-percentile threshold, floor 600, rate guardrail 5223
High-volume clicked domain (`high_volume_domain`)	2238	99th-percentile threshold, floor 200, rate guardrail 2238
Repeated query (`repeat_query`)	149	99th-percentile threshold, floor 12, rate guardrail 149
Repeated query/domain pair (`repeat_query_domain`)	37	99th-percentile threshold, floor 4, rate guardrail 37
Reused exact time-to-click (`reused_ttc`)	40	99th-percentile threshold, floor 40
Same-second click burst (`same_second_burst`)	6	99th-percentile threshold, floor 4

In this run:

- heuristic-only flag rate: 1.36%
- ML agreement-tail reference rate: 2.50%
- operational confidence estimate: 74.49%